Bangzhe Huang

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Research interests: automated research, research world models, AI4AI, scientific machine learning, and many-body physics

Education

Fudan University, Department of Physics Shanghai, China
B.Sc. in Physics, incoming undergraduate *September 2026 – expected June 2030*

Research Experience

College of AI, Tsinghua University Beijing, China
Research Intern, Prof. Ziming Liu's group *Current*

AI4AI: Learning to Design a Meta-model
Independent research; active pilot

- Built a controlled AI4AI pilot in which one meta-model predicts training dynamics and a second model learns architecture-dependent test log-MAE to guide the design of the first model.
- Defined an exhaustive DAG search space from four homogeneous MLP blocks and one softmax gate, generated 10,006 canonical candidates, and built a graph encoder for supervised architecture-to-outcome prediction.
- Implemented the candidate-training and meta-model pipelines and an interactive interface for manual DAG construction, outcome prediction, and automatic architecture design. Preliminary experiments surface a ResNet-like design.

Physics-Informed Many-Body Foundation Models: Scalable Optimization across Heterogeneous Tasks
Project lead; advisor: Prof. Jing Wang, Fudan University; preprint in preparation

- Proposed a task-modular many-body foundation model that preserves each Hamiltonian family's physical priors and ansatzes while learning high-level representations for cross-task composition.
- Led problem formulation, PyTorch implementation, and experimental evaluation against unified joint-training baselines across task scales, task combinations, and fine-tuning settings.
- Preliminary results show comparable generalization and adaptation when unified training is optimizable, while the modular system remains trainable in regimes where the unified baseline becomes ill-conditioned.

Open-Ended Scientific Research Benchmark for Physics Research Agents
Independent research; academic discussion with Prof. Siheng Chen, Shanghai Jiao Tong University

- Designed open-ended physics research tasks with stable ground truth defined by low-dimensional physical structure and equivalent observables, without prescribing a single valid research trajectory.
- Built an end-to-end pilot covering task interpretation, simulation design, code generation and execution, feedback-driven action selection, and final physical auditing.

Phase Transport and Defect Dynamics in Nonreciprocal Spin Systems
Lead researcher; advisor: Prof. Zi Cai, Shanghai Jiao Tong University

- Used Monte Carlo simulations to study the microscopic origin of macroscopic oscillations in two-dimensional nonreciprocal spin systems and extracted oscillation frequency, domain-wall density, and defect observables.
- Proposed the effective transport relation $\omega = \rho v_{\mathrm{eff}}$ and identified two dynamical regimes associated with periodic wall expansion and vortex-constrained topological structures.
- Tested the resulting physical picture across lattice structures and symmetric bond disorder.

Technical Skills

Machine learning: PyTorch, deep learning, scientific foundation models, graph neural networks, LLM agents, benchmark design
Scientific computing: Python, Monte Carlo simulation, numerical experiments, statistical analysis
Physics: quantum mechanics, statistical physics, many-body physics